

Student's Assessment Number.....

THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE, REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MOROGORO REGION FORM TWO TERMINAL EXAMINATION



032

CHEMISTRY

TIME: 2:30Hrs

YEAR: 2026

Instructions

1. This paper consists of sections A, B and C with a total of **ten (10)** questions.
2. Answer all questions in space provided.
3. Section A carries **fifteen (15)** marks, section B carries **seventy (70)** marks and section C carries **fifteen (15)** marks.
4. Calculators, cellular phones and any unauthorized materials are not allowed in the assessment room.
5. All writings must be in black or blue ink except diagrams, which must be in pencil.
6. Write your Assessment Number at the top right corner of every page.
7. The following constants may be used.
Relative atomic masses: H = 1, C = 12, Cl = 35.5, Ar = 40, K = 39, Al = 27, P = 31

For Assessor's Use Only		
QUESTION NUMBER	SCORE	INITIALS
1.		
2.		
3.		
4.		
5.		
6.	GEPAM science hub WhatsApp 0748978398	
7.		
8.		
9.		
10.		
TOTAL		

SECTION A (15 MARKS).

1. For each of the items (i) – (x), choose the correct answer from the given alternatives and write its letter in the table provided below.
 - i. John after finishing to cook he switched off the gas stove but he did not close properly the tap. Later on, he ignites gas stove by using matchbox, suddenly fire outbreak occurred and spreads all over the kitchen. Classify the type of fire caused by John.

A. Class B
B. Class C

C. Class D
D. Class A
 - ii. Form two students were comparing three elements lithium, sodium, and potassium, during the class assignment and finally they realized that these elements belongs to the same group of the periodic table. Which comment do you suggest about these elements?

A. The elements have equal number of electrons in the outer most shell
B. The elements have equal number of shell
C. The elements have equal number of electrons in the inner most shell
D. The elements have equal number of protons.
 - iii. Form one student was asked to remind his fellow students on the laboratory rules during practical session which rule will he insist to be followed?

A. Do not use dirty, cracked or broken apparatus
B. Do not enter in the laboratory without permission of your teach
C. Clean the working surface, benches , and sinks
D. Dress appropriately for the laboratory activities.
 - iv. Mr Mkazi went to the hospital to see a doctor due to the sickness after listening him, a doctor requested full blood picture from the medical laboratory technician. A sample of blood about 10mls was taken from Mr Mkazi left inner elbow bicep. The sample was inserted into a centrifuge machine and all component were identified within a short time. Which branch of chemistry was applied by a laboratory technician?

A. Organic chemistry
B. Biochemistry
C. Chemical engineering
D. Analytical chemistry
 - v. Most of the syrup has the prescription "shake before use" what does this statement implies?

A. It is a clear solution
B. It is a suspension

- C. It is homogeneous mixture
D. It is a filtrate
- vi. X and Y are atoms of elements in the same period of the periodic table. X is in group I and Y is in group II. Which of the following is not true about the X and Y?
A. Y has one more proton than X in its nucleus.
B. The atomic number of Y is one unit greater than that of X

C. X has one electrons less than Y in its valence shell
D. X contain one more electrons than Y in its valence shell
- vii. Mr Karukenge wanted to boil the mixture of glucose and water during chemistry practical session which set of equipment will be suitable for his task.
A. Retort stand, mortar and Bunsen burner
B. Beaker, Bunsen burner and tripod stand
C. Mortor, tripod stand and Bunsen burner
D. Retort stand, test tube and funnel.
- viii. Two boys were arguing about the substances formed after heating calcium carbonate strongly. As a chemistry expert which set of substances will you choose to end up their argument?
A. Metal oxide and gas
B. Metal carbonates and gas
C. Metal hydroxide and gas
D. Metal oxide and water
- ix. The hypothetical element M and N are found in-group VI and II of the periodic table. What is the correct formula for the compound formed between the two elements?
A. MN
B. NM
C. M_6N_2
D. M_2N_6
- x. Which relationship supports the existence of water in three states of matter?
A. Ice → Gas → solid, Solid → Gas → Ice
B. Liquid → Ice → Gas, Gas → Ice → Liquid.
C. Gas → Liquid → Solid, Solid → Liquid → Gas
D. Ice → Gas → Liquid, Liquid → Gas → Ice
2. Match the elements in list A with the number of neutrons in List B, by writing the correct response beside the corresponding item number in the Table provided;

LIST A	LIST B
i. Chlorine	A. Nineteen
ii. Aluminum	B. Twenty
iii. Potassium	C. Sixteen
iv. Phosphorus	D. Fourteen

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v. Argon	E. Eighteen F. Seventeen G. Thirteen H. Fifteen I. Twenty two
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ANSWERS

List A	i.	ii.	iii.	iv.	v.
List B					

SECTION B (70 Marks)

3. (a) A student was given sample of common salt, iron fillings and sand. After performing simple test, he concluded that common salt is a compound. Using your knowledge of compounds, explain how the student can justify that sodium chloride is a compound. (Give four points).

- (b) During a science practical, learners were provided with tea and sugar and asked to differentiate them based on their knowledge about compound and mixtures. As a competent student in chemistry, help them to attempt this question. (Giving four points).

4. (a) During a laboratory experiment students were required to choose a suitable heat source for heating a chemical substance. The teacher recommended using a Bunsen burner. Based on your knowledge of heat sources, explain four reasons why a Bunsen burner is preferable of heat in the laboratory.

(b) In a situation where a Bunsen burner was not available in the school, students were asked to continue with heating experiments using alternative sources. Suggest four alternative sources of heat that can be used in the laboratory apart from a Bunsen burner.

5. (a) A form two student were given the following mixtures and asked to obtain pure substance from each mixture. Identify the methods that can be used to separate the below mixture.

- i. Sand and water_____
- ii. Ethanol and water _____
- iii. Oil from seed_____
- iv. Sodium chloride and water_____
- v. Iodine and Sand_____
- vi. Cooking oil and water_____

(b) Briefly explain how chromatography can be applied in the following fields.

- i. Medicines_____
 - ii. Health and environment
 - iii. Chemistry
-
-
-

iv. Security_____

6. (a) What do you understand by the following terms?

i. Valency

ii. Isotopy

(b) Element W and Z have atomic number 12 and 17 respectively. Use the two elements to answer the following questions;

i. Write the electronic configuration of Z.

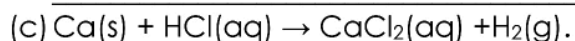
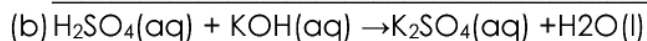
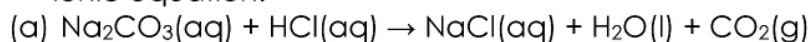
ii. Write the chemical formula of the compound formed when W and Z combine.

iii. What is the valency of W

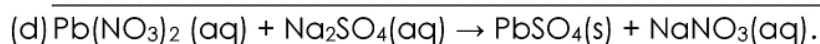
iv. Giving reason predict the type of bond formed by the combination of W and Z.

v.

7. Balance each of the following chemical equations and write its net ionic equation.



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8. (a) Write the chemical symbols of the following elements.

- i. Gold _____
- ii. Copper _____
- iii. Sodium _____
- iv. Potassium _____
- v. Tin _____
- vi. Iron _____
- vii. Silver _____

(b) The language communication among chemists is by using chemical symbols. By using three reasons, explain the advantages of using chemical symbols in chemistry.

9. Study the following part of the periodic table and then answer the questions that follows;

	i						viii
1							
2		ii	iii	iv	v	iv	vii
3							
4							

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- (a) Place the elements having electrons number 1,10,14,16 and 20 in the periodic table by using letter L, M, N, O and P respectively.
- (b) Identify the element which:
- (i) Has the highest electronegativity
 - (ii) Has the valency of four.
 - (iii) Is among the inert gases.
 - (iv) Belongs to alkaline earth metals block.
 - (v) Burn in oxygen to form water.

SECTION C (15 Marks)

10. Describe six applications of neutralization reaction in our daily life.

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